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#include <bits/stdc++.h>
using namespace std;
#define N 4
int minCost = INT_MAX;
bool visited[N];

void tsp(int cost[N][N], int curr, int count, int currCost) {
    if (count == N && cost[curr][0] != 0) {
        minCost = min(minCost, currCost + cost[curr][0]);
        return;
    }

    for (int i = 0; i < N; i++) {
        if (!visited[i] && cost[curr][i] != 0) {

            // Branch and Bound condition
            if (currCost + cost[curr][i] < minCost) {
                visited[i] = true;

                tsp(cost, i, count + 1, currCost + cost[curr][i]);

                visited[i] = false;
            }
        }
    }
}

int main() {
    int cost[N][N] = {
        {0, 10, 15, 20},
        {10, 0, 35, 25},
        {15, 35, 0, 30},
        {20, 25, 30, 0}
    };

    visited[0] = true;

    tsp(cost, 0, 1, 0);

    cout << "Minimum Cost: " << minCost;

    return 0;
}

```

Output

Minimum Cost: 80

=== Code Execution Successful ===